

CMPE 260 – Homework 1

Syntax and Semantics

Student Name: _____

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Question 1 BNF and Context-Free Grammar Construction

Language: Strings of the form $a^n b^m c^n$ where $n \geq 1, m \geq 1$.

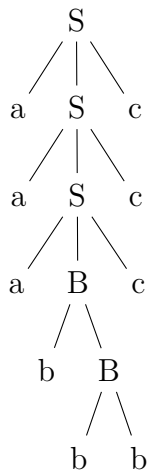
BNF Grammar

$$\langle S \rangle ::= a \langle S \rangle c \mid \langle B \rangle$$
$$\langle B \rangle ::= b \langle B \rangle \mid b$$

Unambiguity

The recursive structure ensures matching numbers of a and c , and bs are isolated. Hence, the grammar is unambiguous.

Parse Tree for aaabbccc



Language Class

Context-Free Language (CFL)

Question 2 Ambiguity in Grammars

Given Grammar

$$E ::= E + E \mid E * E \mid id$$

Ambiguity for $id + id * id$

Two different parse trees:

- $E \rightarrow E + E \rightarrow id + (id * id)$
- $E \rightarrow E * E \rightarrow (id + id) * id$

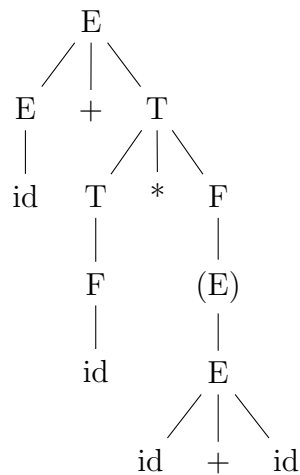
Unambiguous Grammar

$$E ::= E + T \mid E - T \mid T$$

$$T ::= T * F \mid T / F \mid F$$

$$F ::= (E) \mid id$$

Parse Tree for $id + id * (id + id)$



Unambiguity

Operator precedence and left associativity are explicitly defined.

Question 3 Leftmost and Rightmost Derivations

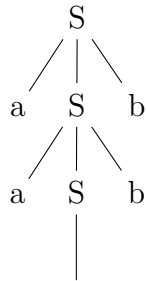
Grammar: $S \rightarrow aSb \mid \epsilon$

Leftmost Derivation of aabb

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aa\epsilon bb = aabb$$

Rightmost Derivation

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aa\epsilon bb = aabb$$

Parse Tree**Language Type**

Context-Free. Cannot be expressed by a regular grammar.

Question 4 Attribute Grammars**Base Grammar**

$$E ::= E + T \mid E * T \mid (E) \mid int$$

Synthesized Attributes

- $E.val = E_1.val + T.val$ for $E \rightarrow E_1 + T$
- $E.val = E_1.val * T.val$ for $E \rightarrow E_1 * T$
- $E.val = int.value$ for $E \rightarrow int$

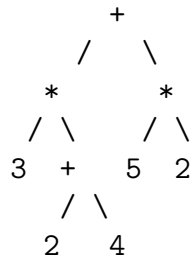
Evaluate: $3 * (2 + 4) + 5 * 2$

$$3 * 6 + 10 = 18 + 10 = \boxed{28}$$

Type Rule and Promotion

Operations must be between operands of same type. If types differ: promote int to float.

Attribute Propagation Example



Question 5 Operational Semantics

Code

```

x = 5
while (x > 0) {
    x = x - 2
}

```

State Transitions

$$x = 5 \rightarrow 3 \rightarrow 1 \rightarrow -1$$

Final Value

x = -1

Modified Condition

```
while (x > 0 && x != 1)
```

Now: $x = 5 \rightarrow 3 \rightarrow 1 \Rightarrow \text{Exit}$

Final x: 1

Question 6 Denotational Semantics

Simple Conditional

```

if (x > 0)
    x = x + 1
else
    x = x - 1

```

Semantic Function

$$[[\text{if } x > 0 \text{ then } x := x + 1 \text{ else } x := x - 1]]\sigma = \begin{cases} \sigma[x \mapsto \sigma(x) + 1] & \text{if } \sigma(x) > 0 \\ \sigma[x \mapsto \sigma(x) - 1] & \text{otherwise} \end{cases}$$

Nested Conditional

```

if (x > 0)
  if (x < 10)
    x = x + 1
  else
    x = x - 1
else
  x = x - 1

```

Recursive Function

```

def update(x):
  if x > 0:
    if x < 10:
      return x + 1
    else:
      return x - 1
  else:
    return x - 1

```

Question 7 Axiomatic Semantics

Code

```

x = x + 2
while (x > 0) {
  if (x > 5)
    x = x - 2
  else
    x = x - 1
}

```

Hoare Triple

$$\{true\} x = x + 2; \text{ while}(x > 0)\{\dots\} \{x \leq 0\}$$

Correctness

- x strictly decreases - Loop terminates - Final $x \leq 0$

Modified Program with y

```
x = x + 2
y = 10
while (x > 0 && y > 0) {
    x = x - 1
    y = y - 2
}
```

Termination

Both x and y decrease; loop terminates when either ≤ 0 .